

z Test for a Proportion

Many hypothesis testing situations involve proportions. A hypothesis test involving a population proportion can be considered as a binomial experiment when there are only two outcomes and the probability of success does not change from trial to trial.

For a binomial random variable $E(X) = np$ and $V(X) = np(1 - p)$.

Since the normal distribution can be used to approximate the binomial distribution when $np \geq 5$ and $n(1 - p) \geq 5$, the standard normal distribution can be used to test hypotheses for proportions.

Test statistic for proportions:

$$z = \frac{\hat{p} - p}{\sqrt{p(1 - p)/n}} \quad (1)$$

where $\hat{p} = \frac{X}{n}$ (sample proportion),

p is the population proportion, and n is the sample size.

1. An educator estimates that the dropout rate for seniors at high schools in Nebraska is 15%. Last year, 38 seniors from a random sample of 200 Nebraska seniors withdrew. At $\alpha = 0.05$, is there enough evidence to reject the educator's claim?
 - (a) State the hypotheses.
 $H_0 : p = 0.15$ and $H_1 : p \neq 0.15$
 - (b) Check whether we can use the standard normal distribution to test the hypothesis.
 $np = 38 \times 0.15 = 5.7$ and $n(1 - p) = 38 \times 0.85 = 32.3$
 - (c) Find the critical value(s). Since $\alpha = 0.05$ and the test is two-tailed, the critical values are 1.96 and -1.96.
 - (d) Compute the test statistic. First, it is necessary to find $\hat{p}$.
 $\hat{p} = \frac{X}{n} = \frac{38}{200} = 0.19$ and $p = 0.15$, so $1 - p = 0.85$
 $z = \frac{\hat{p} - p}{\sqrt{p(1 - p)/n}} = \frac{0.19 - 0.15}{\sqrt{(0.15)(0.85)/200}} = 1.58$
 - (e) Make the decision. We fail to reject the null hypothesis because the test statistic falls outside of the rejection region.
 - (f) Make a conclusion. There is not enough evidence to reject the claim that the dropout rate for seniors in high schools in Nebraska is 15%.

2. An attorney claims that more than 25% of all lawyers advertise. A sample of 200 lawyers in a certain city showed that 63 had used some form of advertising, At $\alpha = 0.05$, is there enough evidence to support the attorney's claim? Use the p-value method.

(a) State the hypotheses.

$H_0 :$ and $H_1 :$

(b) $\hat{p} =$

$p =$ $1 - p =$

$z =$

(c) Find the p-value.

(d) Make a decision.

(e) Make a conclusion.

A recent survey found that 64.7% of the population own their homes. In a random sample of 150 heads of households, 92 responded that they owned their homes. At the 0.01 level of significance, does that indicate a difference from the national proportion?